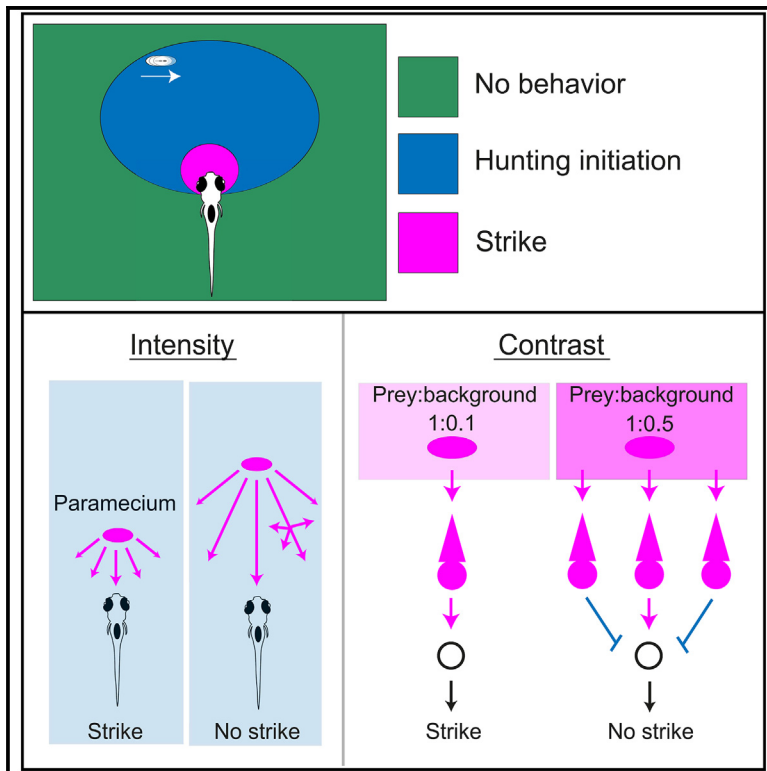


Current Biology

Zebrafish larvae use stimulus intensity and contrast to estimate distance to prey

Graphical abstract



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In brief

Khan et al. show that head-fixed zebrafish larvae can perform prey capture strikes, and this behavior is distance dependent. By manipulating the prey stimuli, they find that stimulus intensity is a key parameter that changes with distance. Stimulus contrast is a second cue that allows for illumination-invariant distance estimation.

Highlights

- Head-fixed zebrafish larvae can perform prey capture strikes
- Strikes and hunting initiation are distance dependent
- Intensity of the stimulus is an important distance cue
- Detecting stimulus contrast allows for illumination-invariant distance estimation



Article

Zebrafish larvae use stimulus intensity and contrast to estimate distance to prey

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SUMMARY

The ability to determine the distance to objects is an important feature of most visual systems, but little is known about the neuronal mechanisms for distance estimation. Larval zebrafish execute different visual behaviors depending on distance; at medium distances, they converge their eyes and approach, but when the prey is close enough, they execute a strike and suck the prey into their mouths. To study distance estimation, we developed a head-fixed strike assay. We found that we could evoke strike behavior in head-fixed larvae and quantify head movements to classify the behavior as a strike. Strikes were dependent on distance to prey, allowing us to use them to study distance estimation. Light intensity is rapidly attenuated as it travels through water, so we hypothesized that larvae could use intensity as a distance cue. We found that increasing stimulus intensity could cause larvae to strike at prey that would normally be out of range, and decreasing the intensity could lower the strike rate even for very proximal stimuli. In addition, stimulus contrast is a key parameter, and this could allow larvae to estimate distance over the range of natural illumination. Finally, we presented prey in the binocular vs. monocular visual field and found that monocular prey did evoke strikes, although the binocular input produced more. These results suggest that strike behavior is optimally evoked by bright UV dots in the binocular zone with minimal UV background light and provide a foundation to study the neuronal mechanisms of distance estimation.

INTRODUCTION

Prey capture behavior in larval zebrafish is a multi-step process evoked by small moving objects such as paramecia. Upon spotting a prey object, larvae first converge their eyes and use tail movements to turn toward and approach the prey^{1,2} before finally consuming it. The strike is the final phase of prey capture, during which they use suction to draw the paramecium into their mouth. The suction force is generated by the sudden expansion of the oral cavity via a fast downward movement of the jaw and upward movement of the head.^{3,4} The strike can be accompanied by swim bout or occur without any tail movement,⁵ and the accompanying swims are kinematically distinct from other behaviors.^{4,6}

Both strikes and prey capture initiation are distance dependent; prey capture initiation can occur when the larva is several mm away from the prey, but the strike is not evoked unless the larva is within 2 mm of the prey or close enough to consume it.⁵ Accurately judging distance before executing a strike is important to ensure that the larva is directly below the prey

when suction is applied. The fact that the strike is typically only evoked when the prey is close to the larva suggests that larvae must have some mechanism to estimate distance, but the nature of the visual cues for distance estimation in zebrafish larvae is unknown.

It is possible that larvae use monocular cues, binocular cues, or both to estimate distance and decide whether prey is within striking distance. Binocular cues have long been thought to play a role in prey capture, as the defining feature of hunting initiation is eye convergence, which creates a binocular zone in the central visual field.⁷ However, input from one eye is sufficient for larvae to successfully perform strikes and capture prey, albeit at a lower rate.⁴ Resolution of this question has been challenging, in part because the behavior could only be studied in freely swimming larvae, making it difficult to present controlled visual stimuli to test the nature of the cues.

Here, we develop a head-fixed preparation that allows us to observe strike behavior in response to nearby prey and an algorithm to detect strikes. The initial phases of prey capture behavior, including eye convergence, orienting, and approach



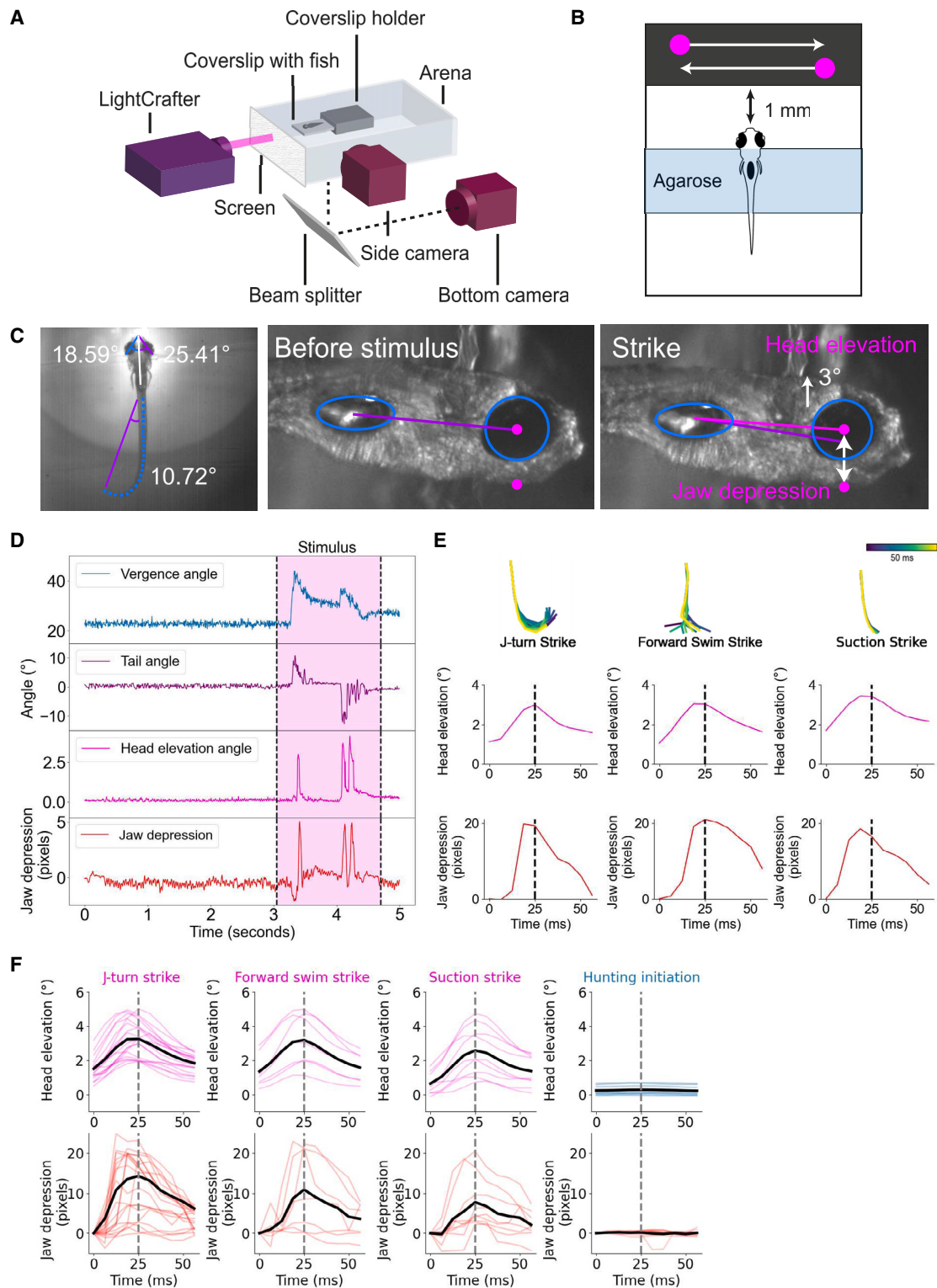


Figure 1. Head-fixed strike behavior can be induced with artificial prey

(A) Schematic of the behavioral setup showing side and bottom cameras and projection of prey onto one wall of the arena.

(B) Schematic of prey stimulus and distance to screen.

(C) Left, bottom camera view showing eye and tail angle tracking. Middle and right, side camera view before and during a strike. Eye to swim bladder angle was used to detect head elevation, and eye to hyoid bone distance was used to detect jaw depression.

(legend continued on next page)

swims, have been studied in a head-fixed preparation,^{7,8} but so far, head-fixed strikes have not been described. This paradigm allowed us to characterize the visual stimuli that best evoke strikes and show that they are distance dependent. We then manipulated our prey stimuli to probe the mechanism of distance estimation and found that the intensity and contrast of the stimulus are key parameters to determine distance to prey.

RESULTS

Head-fixed larvae perform strikes

We modified our head-fixed prey capture assay⁸ by removing agarose around the eyes and mouth and recording head movements with a high-speed camera positioned on one side of the larva, in addition to the camera that records eye and tail movements (Figure 1A). We also used a spatial visual stimulator with UV, blue, green, and red channels (Figures S1A and S1B)⁹ to present a 4° diameter UV dot that swept horizontally across the screen at 200°/s. With the larva 1 mm from the screen (Figure 1B), we found that a UV prey stimulus could evoke behavior with similar kinematics to a capture strike. In addition to the previously described eye and tail movements of prey capture,^{7,8} head-fixed larvae also performed movements that appeared similar to the capture strike, including head elevation and depression of the hyoid bone of the jaw (Video S1).^{3,4} We also observed trials where hunting initiation (eye convergence and tail movements) occurred, but the putative strike was absent (Video S2). We quantified the change in head elevation angle by using the eye and swim bladder as landmarks and also tracked jaw depression relative to the eye (Figure 1C, right). We found that after hunting initiation (eye convergence and tail motion onset), we could see distinct spikes in head elevation, which occurred simultaneously with jaw depression (Figure 1D). As the upper part of the head rapidly tilts up, the jaw moves down, which results in little absolute displacement of the jaw, but its distance from the eye is increased (Video S1; Figure 1D). Rapid jaw depression tends to specifically occur during putative strikes, as we can see by plotting the maximum jaw depression derivative during trials with and without head elevation (Figure S2A). The time between maximal head elevation and jaw depression was usually around 10 ms, whereas hunting initiation was not similarly coincident with jaw depression (Figure S2B). These head and jaw movements occurred over a time frame of 25 ms (Figure 1E), similar to the kinematics observed in free-swimming larvae.³

The tail movements that we observed during the strikes were mostly J-turns (Figure 1E, first column), whereas in free-swimming fish, strike tail kinematics typically consist of forward swims⁴ or an absence of movement in the case of suction strikes.⁵ This discrepancy may be due to some differences in the prey stimulus in the two conditions. For free-swimming larvae, after the initial orienting swims, the prey stimulus is typically located near the center of the visual field by the time the strike occurs, while in our preparation, the UV dot was moving

much more rapidly and thus was not centered. We therefore designed another stimulus in which a dot swept to the center of the visual field and remained there, flickering, to mimic the natural condition. This stimulus had a lower efficacy in inducing hunting (~50%) but did evoke other types of strikes, including forward swims and suction strikes (Figure 1E, middle and right column, Figures S2C and S2D; Videos S3 and S4). These strikes were also characterized by simultaneous head elevation and jaw depression, occurring on a similar time scale (Figure 1E, rows 2 and 3, and population data, Figure 1F), whereas hunting initiation was not accompanied by these head and jaw movements. Thus, we are able to evoke the distinctive strike head and jaw movements that are intended to suck prey into the larva's mouth, although in our preparation, the tail movements are often not identical, probably due to differences in the stimulus.

Quantitative classification of strikes

The maximum angle of head elevation achieved during a strike can vary from fish to fish in our preparation, so we developed an algorithm to simply quantify the total upward movement of the head as a way to identify strikes. We first identified frames with motion (see STAR Methods) and then used the dense optical flow Farneback algorithm¹⁰ to calculate the magnitude and direction of motion of each pixel in the motion frames. We found that although non-strike prey capture and strike frames both have moving pixels, in strike frames, there was an overrepresentation of upward-moving pixels compared with the three other directions (Figure 2A). The same held true in a dataset with eye convergence and strike trials from ten larvae; plotting the number of pixels for each motion direction over time, we saw an increase in all motion directions during hunting initiation without strikes (Figure 2B), but for strike trials, we see many pixels moving up as a result of head elevation (Figure 2C). Therefore, we defined strikes as motion frames where the most frequent pixel direction (mode) was upward. Plotting the timing of strikes in response to a relatively slow stimulus (60°/s) shows that strikes are tightly coupled to the presentation of prey stimuli and only occur when the stimulus is within 30° of center, with the largest number happening at -10° (Figures 2D and 2E). In contrast, hunting initiation can occur when the stimulus is slightly more lateral (Figure 2E). We quantified the accuracy of the algorithm by comparing its results with human annotation. We found that we were able to correctly identify 92% of the strikes, with a low rate of false positives (Figure 2F). The few false positives we did observe were due to struggles, which can include large upward movements. However, these were relatively rare during prey presentation, and the vast majority of frames with upward motion in our dataset occurred during strikes.

Characterizing the optimal strike stimulus

The stimulus to trigger strikes has never been characterized, so we first explored the basic stimulus parameters to identify the optimal strike stimulus, as we previously did for hunting.⁸ Here, we defined hunting initiation based on eye convergence, and

(D) Time course of hunting onset, head elevation, and jaw depression in one trial.

(E) Examples illustrating the three types of tail movements observed during strikes and the corresponding head elevation and jaw depression time course. Dotted line represents peak head elevation.

(F) Head elevation and jaw depression during the three types of strike and eye convergence onset. See also Figure S1 and Videos S1, S2, S3, and S4.

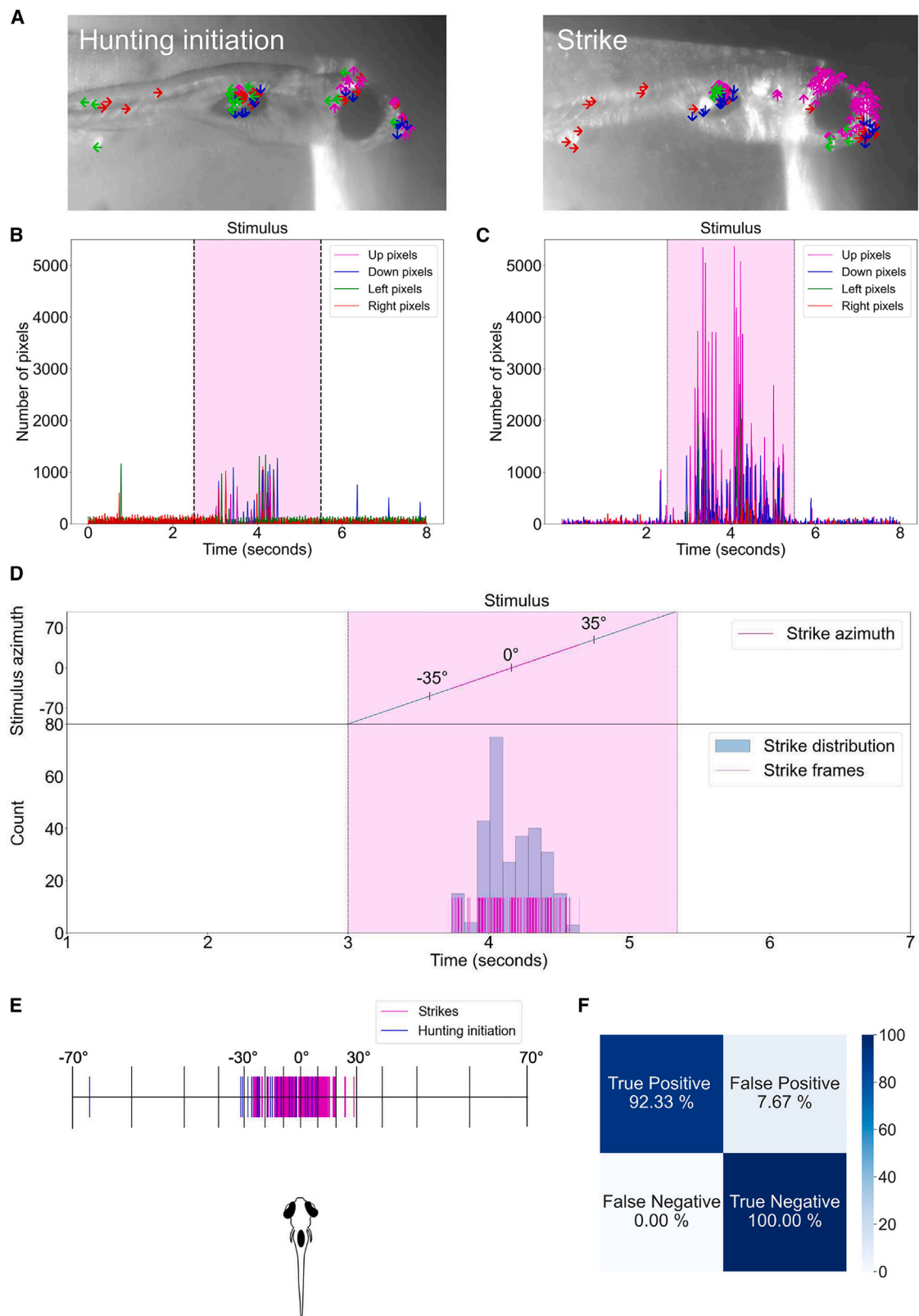


Figure 2. Head elevation can be quantified and used to identify strikes

(A) Moving pixels during hunting initiation and a strike, represented by arrows color coded for direction. Magenta arrows represent upward motion.
(B) The number of pixels moving in the four directions over time in ten non-strike hunting trials in ten different larvae.

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eye convergence rate was the percentage of trials that included an eye convergence event. Similarly, strike rate was the percentage of trials that had at least one strike according to our algorithm. We first characterized the optimal size and speed to evoke strikes when the screen was 1.5 mm from the larva. We varied the size of the stimulus from 1° to 18° to determine the ideal size and found that a stimulus between 4° and 8° was ideal to trigger strikes, with a strike rate of over 80% (Figure S3A). This is similar to the size range that triggered hunting initiation. We then varied the speed of the 4° dot and found that stimuli moving at a speed of 60–200°/s gave us the highest strike rate, similar to the speeds that evoke hunting initiation optimally (Figure S3B).

We next asked whether other wavelengths of light could evoke strikes and prey capture as effectively as our UV stimulus. We found that, while the UV prey stimulus evoked eye convergence in 90% of trials, the rate was 20%–30% for blue and green stimuli, respectively, and around 10% for red stimulus (Figure 3A). The decrease in strike rate was even more dramatic, going from 85% for UV prey to 10% for blue and green stimuli and effectively zero for red (Figure 3B). This is consistent with recent studies showing that UV light is more effective than visible light in inducing hunting initiation.¹¹

The head-fixed strike is distance dependent

To test whether the strike behavior we observed is distance dependent, we moved the larva to varying distances from the screen (Figure 4A) and recorded behavior while presenting the same prey stimulus (Figure 4B). Size and speed of the stimulus were not updated, meaning that from the larva's perspective, it became smaller and slower with increasing distance. We observed that both eye convergence and strike behaviors were much less common when the distance was increased to 7 mm, and the rate further decreased when larvae were 12 mm from the screen, with strike behavior decreasing relatively more than eye convergence (Figure 4C). At 7 and 12 mm, we saw trials that included eye convergence but no strikes, which were not observed at 1.5 mm (Figure 4C). These "hunting initiation only" trials represent cases where the larvae could see the prey but did not try to consume it. This is reminiscent of free-swimming behavior in response to prey at a medium distance, which triggers prey capture initiation but not strikes.⁵ In addition, if we plot the conditional probability of a strike occurring given eye convergence, we can see that as distance increases, this conditional probability decreases, for some fish, quite dramatically (Figure 4D).

Increasing stimulus intensity can compensate for greater distance

One possible reason for the lower number of strikes at 7 and 12 mm is the decrease in stimulus intensity at these further distances. Water scatters and absorbs light more than air, which means that light intensity goes down significantly over relatively short distances,¹² and the larva should experience a dimmer stimulus as the distance increases (Figure 4E). To quantify how

much stimulus intensity changed, we measured the power lost at these distances by placing a power meter sensor underwater in the arena, where the larva would normally be, and projecting the visual stimulus on the screen. We found that the power of our 4° UV stimulus was measured as 20 nW at 1.5 mm, but it decreased to 17 nW at 7 mm (12% attenuation), and for 12 mm, the power went down to 15 nW, so the attenuation was 22% (Figure S4A). We then asked whether compensating for the loss of power over distance could improve the strike rate (Figure 4F). We increased the UV LED power and then measured the intensity of the UV dot to ensure that it was equivalent to the 1.5 mm condition. By increasing the intensity to compensate for distance and keeping all other stimulus parameters constant, we were able to increase the hunting initiation and strike rates to nearly the levels seen with the screen 1.5 mm away (Figure 4G).

As the larvae were moved farther from the screen, the size of the stimulus from the larva's perspective also decreased, so we conducted the experiment with stimulus size adjusted to be consistently 4° and found that the behavior similarly decreased at 7 and 12 mm (Figure S4B), indicating that stimulus intensity is more likely the key parameter affected by screen distance.

The distance range of 12 mm, across which we see a transition from low to high rates of hunting initiation and strikes, is much larger than that previously reported for both free-swimming strikes and prey capture.⁴ This may be due to the fact that a paramecium is an omnidirectional light source, and since its light is scattered in all directions, the intensity of light should decrease quite steeply as it is moved further away, according to the inverse square law (Figure 5A).¹³ As the distance from an omnidirectional light source is doubled, light intensity is divided by 4. In contrast, our projector's light is collimated, and its more directed light should not fall off as quickly. To test whether this difference between natural and artificial prey could account for our greater strike distances, we calculated the expected decrease in intensity for distances from 1 to 4 mm. For a strong UV prey stimulus of 35 nW, the intensity should drop below 5 nW at 3 mm according to the inverse square law. We compared this with the intensity of our projected 35 nW stimulus, which we measured to still have a power of 33 nW at 3 mm (Figure S5). We then asked whether decreasing the projected stimulus intensity at 3 mm according to the inverse square law could recapitulate the shorter strike range observed in free-swimming larvae. We found that, indeed, the strike rate went from nearly 100% at 1 mm to ~20% at 3 mm (Figure 5C), with a decrease in conditional probability of strikes as well (Figure 5D). This indicates that the dramatic fall off of stimulus intensity over short distances makes stimulus intensity a very informative cue for the decision to strike.

We also performed the intensity compensation experiment for 1, 2, and 3 mm and found that we were similarly able to trigger the same amount of behavior at 3 mm as at 1 mm by increasing stimulus intensity to compensate for attenuation (Figure 5E). Thus, UV intensity seems to be an important parameter for

(C) The number of pixels moving in each direction during ten strike trials.

(D) Timing of strikes in response to a dot sweeping at 60°/s from left to right ($n = 5$ larvae).

(E) Position of the stimulus when eye convergence or strikes were evoked.

(F) Confusion matrix of algorithm accuracy compared with human annotation.

See also Figure S2.

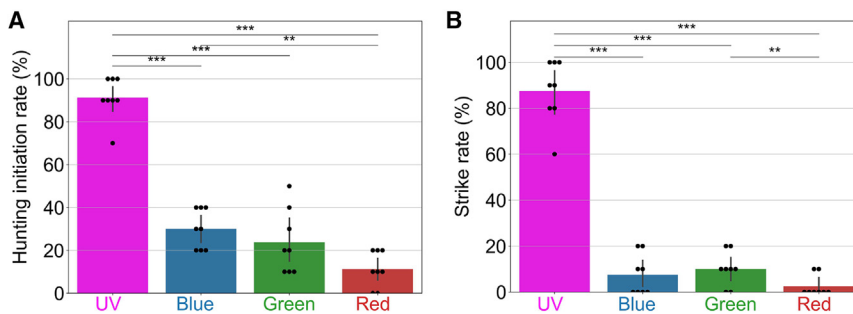


Figure 3. UV prey stimuli evoke the most eye convergences and strikes

(A and B) Hunting initiation and strike rate for the ideal size and speed stimulus using different wavelengths ($n = 8$ larvae). The stimulus power for all wavelengths was 27.20 nW. Black dots represent data from one larva. ** $p \leq 0.01$, *** $p \leq 0.001$, Mann-Whitney U test. Error bar represents SD. See also Figure S3.

distance estimation, as increasing this parameter alone, without changing size or speed, was sufficient to increase strike rate.

We then reasoned that if intensity is a key distance cue, lowering the intensity of the stimulus should cause a decrease in behavior even at close range. We placed the larva 1 mm from the screen and presented prey stimuli with intensities varying from 35 to 2 nW, which would correspond to the decrease in stimulus intensity over 4 mm according to the inverse square law (Figure 5F). We found that hunting initiation and strike rate decreased as intensity was lowered, with strike behavior being more sensitive (Figure 5G), and the conditional probability of strikes also decreased (Figure 5H), suggesting that by simply decreasing the intensity, we can mimic the effect of distance.

Larvae use the ratio of prey to background UV light to gauge distance

Using UV intensity alone to determine distance to the prey could be problematic, as ambient lighting conditions can vary in nature, with cloud cover decreasing irradiance by up to 50%¹⁴ and time of day also affecting light level. Thus, if the putative UV-intensity-sensing circuit was optimized for the brightest light condition, then little to no successful prey capture would occur at other times of day. One strategy to achieve light-level invariance would be to use the ratio of different wavelengths within the prey object as a cue for object proximity. Since UV intensity falls off more quickly in water than blue and green light over a given distance, nearby prey should be UV bright, whereas more distant prey should appear blue/green in color. To verify that UV light is more attenuated, we measured the power over 1.5 to 12 mm and observed a greater attenuation of UV light compared with blue and green wavelengths (Figure S4A). We then asked whether larvae could use the UV-to-blue light ratio within the prey object to estimate distance by adding blue light to the UV prey stimulus itself, starting with a ratio of 1 UV:2 blue, based on the relative power of those wavelengths in the environment¹⁵ and decreasing the UV power according to the expected attenuation over 1–15 mm (Figure S6A). We found that a significant change in the level of hunting initiation and strikes did not occur until the ratio reached 0.7 UV:2 blue, which corresponds to 15 mm (Figures S6B and S6C). Thus, spectral ratio is most likely not playing a role in the decision to strike, although it could be used for prey detection.

Another strategy for light-level invariance could be to use the relative amount of background light to modulate behavior. As the ambient light level increases, the intensity of the prey stimulus and the background should both go up, and although the

intensity of the prey varies with the lighting condition, the ratio of the stimulus to background intensity should be constant for a given distance. Therefore, by also taking into account stimulus contrast, larvae could use intensity as a light-level-invariant distance cue. To test this hypothesis, we generated stimuli with different background intensities relative to the prey; for example, for a medium-intensity prey stimulus of 17 nW, we used background to prey ratio of 0.1 (background power = 1.7 nW/4° circle area) to 0.5 (background power = 8.5 nW/4° circle area) (Figure 6A). We first tested at what point increasing the background UV level would inhibit hunting initiation and strikes for a medium-intensity (17 nW) UV prey stimulus and found that a decrease in behaviors was evident at a ratio of 1:0.1 of prey and background UV power, with strong suppression at 1:0.2 (Figure 6A). To ask whether this property would scale, we chose a low (7 nW) and high (34 nW) prey stimulus intensity and tested whether the addition of the same ratios of UV power in the background would have a similar effect (Figure 6B). We found that hunting initiation and strikes in response to the high- and low-intensity stimuli were also suppressed starting at a background ratio of 0.1, with strike behavior being slightly more affected, and both behaviors were strongly suppressed by a background ratio of 0.2 (Figures 6C–6G). Therefore, the ratio at which background UV starts to affect hunting initiation and strike rate is consistent across a range of intensities, which is an essential condition for this parameter to be used as a lighting-invariant distance cue.

We next asked whether background light of other wavelengths would also suppress behavior. We found that adding a green background of up to three times the UV intensity had no effect on the hunting initiation or strike rate triggered by the high-, medium-, or low-intensity UV prey stimulus (Figures S6E and S6F). We also tested blue and red backgrounds up to three times and found that they similarly had no effect on the hunting initiation and strike rate (Figures S6G and S6H), indicating that the effect is specific to UV wavelengths.

Eye convergence angle and binocular cues may play a role in strike behavior

So far, we have provided evidence for the role of stimulus intensity and contrast in distance estimation, and these are monocular cues. However, it is likely that there is some role for binocular vision in prey capture, as eye convergence to allow binocular viewing of the prey is a signature of larval hunting.^{2,7} In humans, eye convergence itself can be a distance cue, as when both eyes foveate a close target, they will be more converged than for a more distant object.¹⁶ Eye convergence angle could be used

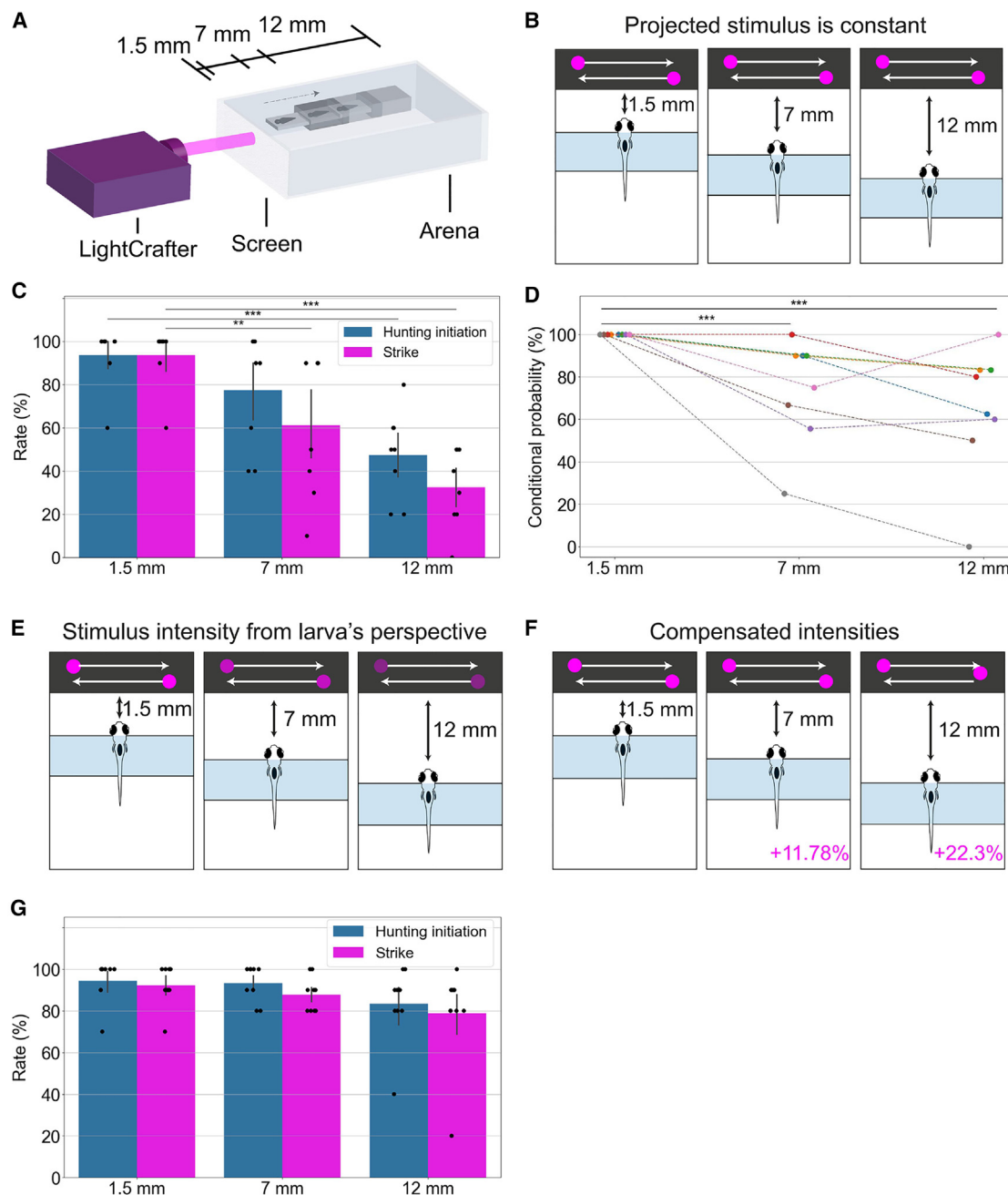


Figure 4. Eye convergence and strike behaviors are distance dependent

(A) Movement of coverslip holder and larva away from the screen to increase the distance to prey.
(B) Identical prey stimuli (power = 20 nW) were projected on the screen, whereas the larva was moved to varying distances from the screen.
(C) Hunting initiation and strike rate decreased as distance to the screen increased ($n = 8$ larvae).
(D) Conditional probability of strikes given eye convergence at three distances; colors represent individual larvae ($n = 8$ larvae).
(E) The stimulus intensity decreases from the larva's perspective as distance is increased.
(F) Increasing the stimulus power by 11.78% for 7 mm and 22.3% for 12 mm to match the 1.5 mm stimulus.
(G) Hunting initiation and strike rate at 1.5, 7, and 12 mm after intensity compensation ($n = 9$ larvae). ** $p \leq 0.01$, *** $p \leq 0.001$, Mann-Whitney U test. Error bar represents SD. See also Figure S4.

for distance estimation in zebrafish, as free-swimming larvae converge their eyes slightly more when prey is closer.⁵ We therefore asked whether there was any increase in convergence angles for shorter distances (data from Figure 5C) and found that

strike trials at different distances had similar maximum convergence angles (Figure S7A). However, by looking at strike trials, we might be selecting trials in which the prey was perceived as relatively close despite the distances. We therefore compared

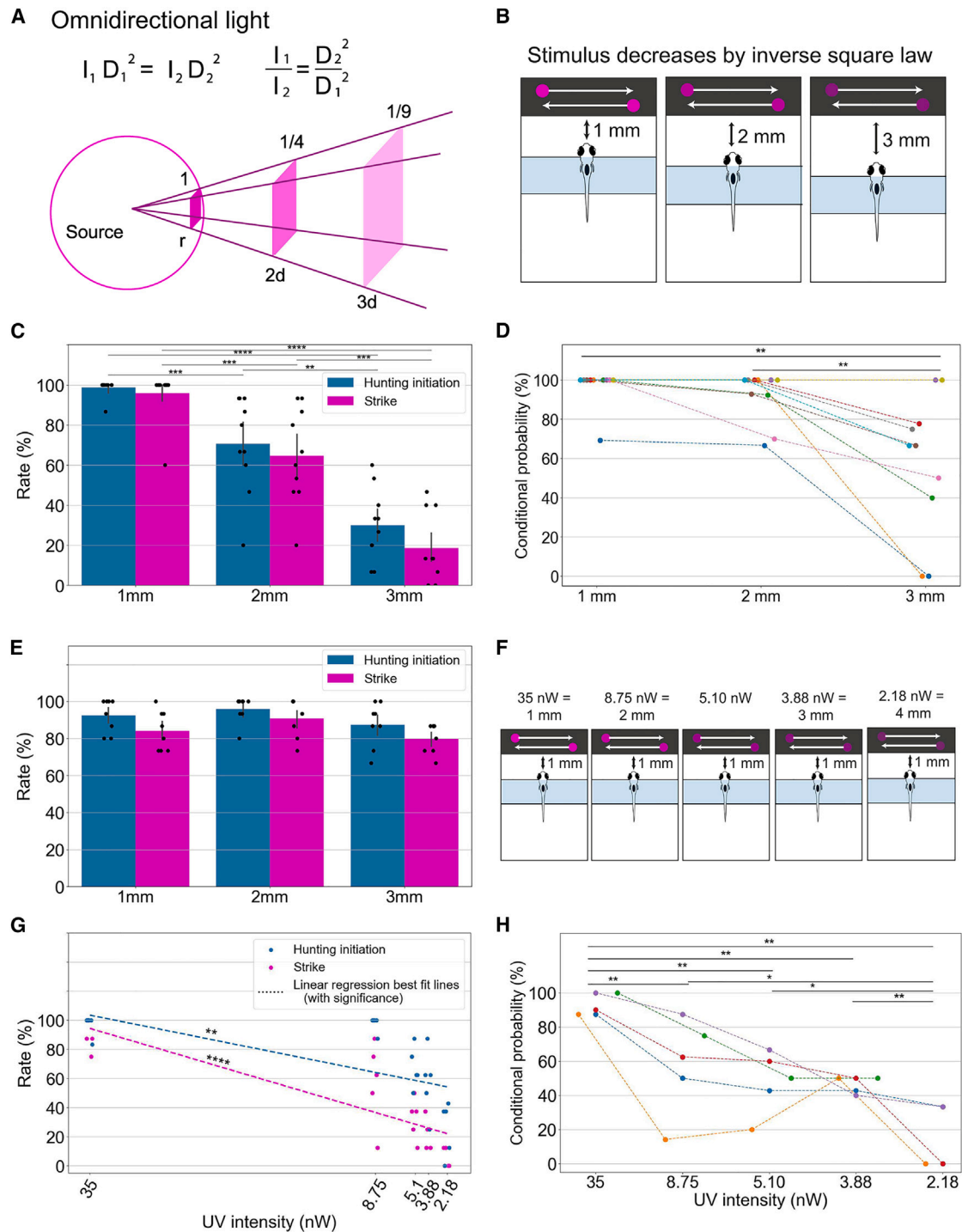


Figure 5. Paramecium intensity should decrease dramatically over 3 mm, and simply lowering the intensity of the stimulus at 1 mm can mimic the effect of distance

(A) Falloff of light intensity as a function of distance for omnidirectional vs. directed light sources, adapted from Johnston and Fauber.¹³

(B) Stimuli at 1–3 mm with power adjusted according to the inverse square law.

(C) Hunting initiation and strike rate in response to the stimuli in (B) ($n = 9$ larvae).

(D) Conditional probability of strikes given hunting initiation for data in (C).

(E) Hunting initiation and strike rate at 1, 2, and 3 mm after intensity compensation ($n = 9$ fish).

(F) Stimuli of different intensities presented with the screen 1 mm away. Power values correspond to stimulus power at 1–4 mm for an omnidirectional light source.

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strike trials to hunting-only trials at the same distance and found that the strike trial convergence angle was larger (Figure S7B). We then asked whether, for the same trial, the eyes were more converged during the strike vs. the moment of hunting onset. For this, we used a relatively slow prey stimulus of 60°/s to maximize the time gap from hunting onset to first strike, and we found that, indeed, the eyes were more converged at the time of the strike (Figure S7C). Thus, it is possible that eye convergence angle plays a role in distance estimation, either in the decision to strike or the finer judgment of whether to do a suction strike,⁵ attack swim, or S-strike.⁴

Finally, we asked whether binocular information is required for hunting initiation and strikes by presenting prey in the binocular vs. monocular regions of the visual field. We calculated that with the screen 1.5 mm away and the eyes maximally converged, the binocular zone should extend about 25° left and right from the center of the visual field (Figures S7D–S7J). To present prey only in the monocular part of one side of the visual field, we modified our prey stimulus to only sweep from an azimuth of –70° to –30° or from 30° to 70° (Figure 7A) and measured hunting initiation and strike rates. We found that the binocular prey stimulus (–30° to 30°) evoked a high percentage of eye convergence and strikes, while the lateral, monocularly presented prey also evoked some hunting initiation and strikes but at a much lower rate (Figure 7B). These results show that visual input to both eyes is not absolutely required to induce strikes, but binocular input does dramatically increase the occurrence of strikes.

DISCUSSION

Head-fixed strike assay

We have shown that prey capture strikes can be evoked in head-fixed zebrafish larvae. Previous experiments on head-fixed larvae have described the initial stages of hunting (eye convergence, J-turns, and forward swims),^{8,17,18} but head-fixed strikes had never been observed. This may be because the head elevation and jaw depression that characterize strikes are most obvious from the side, rather than vertical view. In addition, earlier studies positioned the screen 10–35 mm away from the larva, lowering the probability of evoking strikes. Another reason could be the use of red or blue light rather than UV light. While visible light can evoke some prey capture (in our hands, a 20%–30% prey capture rate for blue and green stimuli), the strike rate was less than 10% (Figure 3), compared with 80%–90% for UV stimuli.

Distance dependence of strike behavior

At what level is the decision to execute hunting initiation and/or strike behavior made? Previously, we identified retinal ganglion cells (RGCs) that respond specifically to prey stimuli, and one possibility is that larvae will initiate hunting if enough of these RGCs are activated or if the activation is strong enough. These RGCs may then activate prey capture command-like neurons in the pretectum,¹⁸ which trigger eye convergence and tail movements. At some point, if the prey is too far away, the stimulus could be

insufficient to activate enough prey-responsive RGCs, or the activation too weak, and no behavior is observed. Strike behavior seems to have a higher threshold than eye convergence, which could be due to a requirement of more RGC activation or a different set of RGCs with more stringent selectivity. There may be a separate set of command neurons activated by nearby prey, which trigger head elevation and jaw depression to consume the prey. In the future, it will be interesting to explore these hypotheses with functional imaging.

Intensity and contrast may be used to gauge distance

Our manipulations of stimulus intensity and background ratio suggest that these are important distance cues. How could differing intensity and contrast modulate neuronal responses? Increasing the intensity of the prey stimulus could increase the number of active prey-responsive RGCs or the level of their activity, while UV background intensity could have an inhibitory effect on this response. The suppression by background could occur via local lateral inhibition or it could be a more global adaptation to the overall light level, both of which could be accomplished at the level of horizontal cells¹⁹ or later stages of processing.^{20,21} Within the zebrafish retina, out of the three types of cone horizontal cells, there is one type, H3, that primarily contacts UV cones²² and could mediate the suppression by background UV levels. It is also possible that this effect could occur after the retina, so it will be interesting to use functional imaging to test whether these phenomena can be observed at the level of the RGCs.

How would intensity and contrast interact in nature? As the overall light level increases, the intensity of the prey stimulus should increase, as well as the background UV level, such that the contrast of a given prey object at a certain distance will be constant, despite the changing lighting conditions. Larvae could respond to low-contrast prey with no behavior, medium-contrast prey with hunting initiation, and high-contrast prey with strikes. This would allow them to respond appropriately with prey capture orienting and approach swims to prey at a medium distance and release strike behavior only once the prey is close enough to be consumed. Both intensity and contrast should be informative distance cues for many aquatic species, so it will be interesting to see whether other animals use similar mechanisms to gauge distance to prey, predators, or conspecifics.

The role of eye convergence and binocular information in the decision to strike

We found that larvae did converge their eyes more in strike trials than in hunting-initiation-only trials, so we cannot exclude the possibility that eye convergence also acts as a distance cue. However, there are a number of other possible interpretations of this result. It could be that the decision to strike and to maximally converge the eyes are both downstream of the same distance cue. Another possibility is that eye convergence is used specifically for fine distance estimation, in the decision to perform a suction strike in response to the most proximal prey,

(G) Hunting initiation and strike rate for prey stimuli of different intensities at 1 mm. (n = 5 larvae). Asterisks denote the p value of the slope, describing the relationship between the intensity and the behavior rate.

(H) Conditional probability of strikes from data in (G). *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001, Mann-Whitney U test. Error bar represents SD. See also Figure S5.

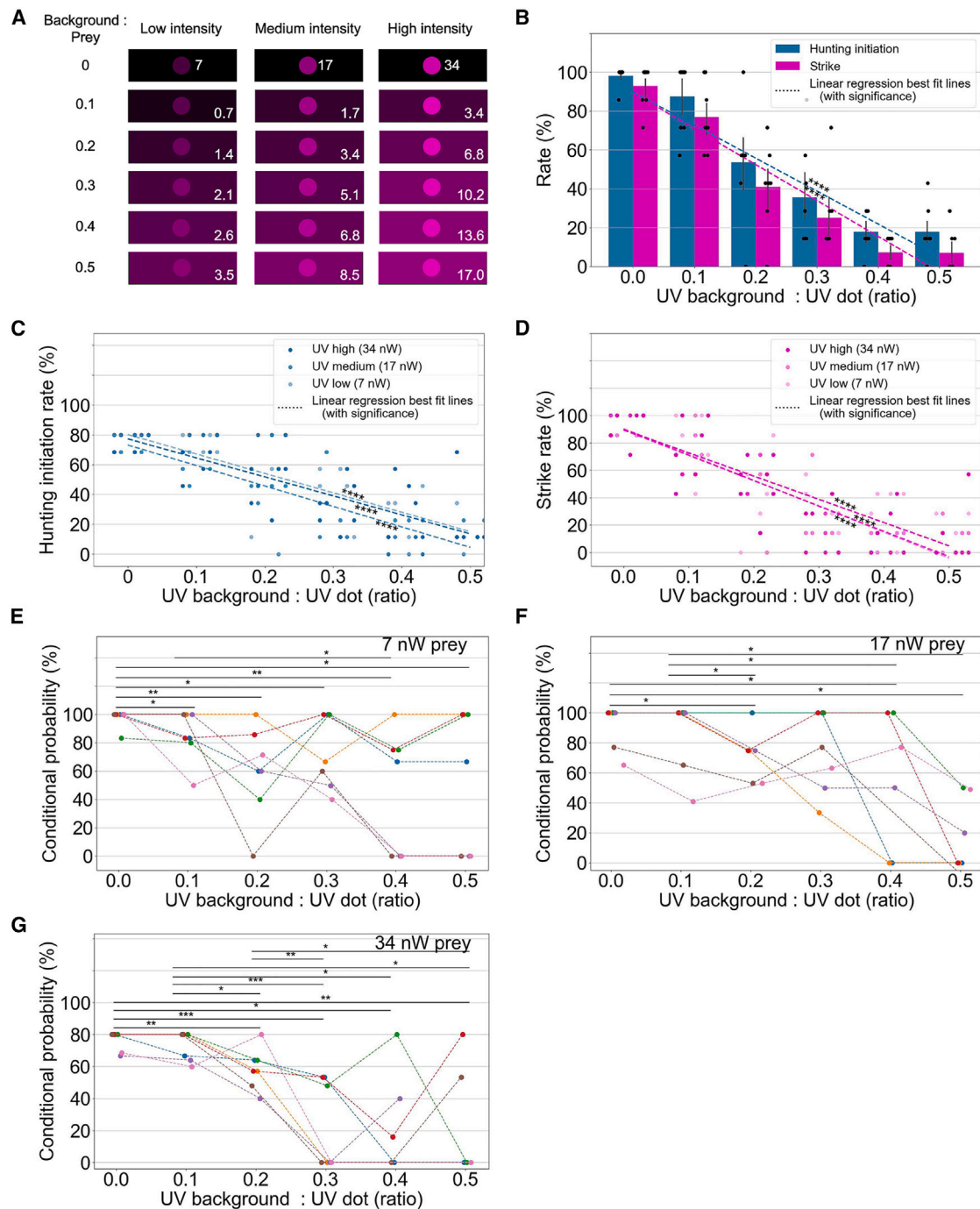


Figure 6. Behavior is suppressed by UV background light at a ratio of 1:0.2, and this scales over different intensities

(A) Stimuli with varying UV prey:background ratios across different prey intensities.

(B) Hunting initiation and strike rate in response to the medium-intensity UV prey (17 nW) with varying background ratios (n = 7 larvae).

(C) Hunting initiation rate for high-, medium-, and low-intensity prey stimuli with varying background UV.

(D) Strike rate for high-, medium-, and low-intensity prey stimuli with varying background UV (n = 7 larvae).

(E) Conditional probability of strikes for a 7 nW prey stimulus with UV background ratio of 0–0.5.

(F) Conditional probability of strikes for a 17 nW prey stimulus with UV background ratio of 0–0.5.

(G) Conditional probability of strikes for a 34 nW prey stimulus with UV background ratio of 0–0.5 (n = 7 larvae). *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001, Mann-Whitney U test. Error bar represents SD.

See also Figure S6.

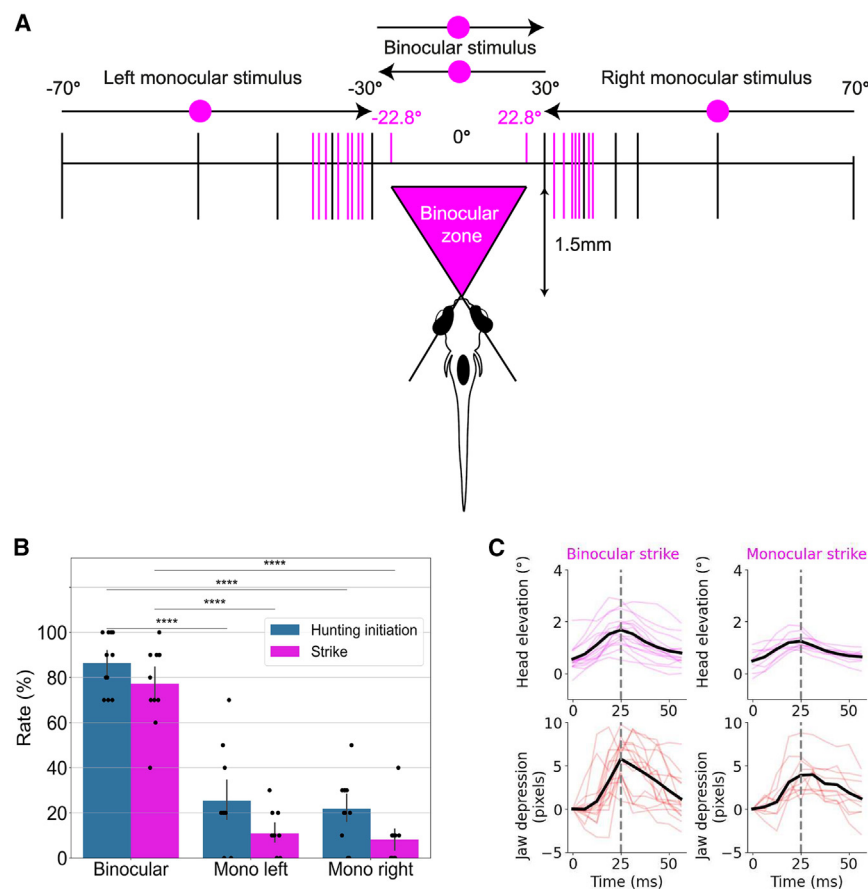


Figure 7. The role of binocular information

(A) Schematic showing the presentation of prey in the monocular zone (-70° to -30° or 30° to 70°) and stimulus position during strike frames (magenta lines).

(B) Hunting initiation and strike rate in response to binocular, left monocular, and right monocular prey ($n = 10$ fish).

(C) Head elevation and jaw depression during binocular and monocular strikes ($n = 10$ larvae). **** $p \leq 0.0001$, Mann-Whitney U test. Error bar represents SD.

See also Figure S7.

vs. attack swims or S-strikes for slightly more distant prey. We argue here that the intensity and contrast of the prey stimulus very strongly modulate the decision to hunt and to strike, but these are not the only visual parameters important for the entire prey capture sequence. In the future, it will be interesting to ask how prey distance affects eye convergence and how essential proper convergence is for the behavior.

We presented monocular and binocular prey stimuli as a simple test of the necessity of binocular input for hunting initiation and strikes (Figures 7B and 7C). One caveat is that we are stimulating different parts of the visual field in the monocular vs. binocular conditions, so part of the difference could be due to the presence of prey in the central visual field in the binocular condition. However, it is interesting to note that the monocular stimulus did evoke strikes, albeit at a low rate. This suggests that the mechanism for triggering strikes does not completely require binocular input, although it may be facilitated by it, and this is consistent with the finding that monocular free-swimming larvae do occasionally perform strikes.⁴

There are multiple ways that binocular information can be useful for visual behaviors; in animals with stereopsis, the binocular disparity, or difference in the viewpoints of the two eyes can give rise to depth perception.²³ A second and separate binocular advantage is that the availability of input from two eyes can increase visual acuity and contrast sensitivity via binocular summation.^{24–26} It is unclear whether the zebrafish visual system has all the prerequisites necessary to for stereopsis, for example,

ensuring that features seen by the two eyes are matched.²⁷ However, we know that larvae have to solve an extremely challenging sensitivity and acuity problem to detect prey. Visual acuity in larval zebrafish is limited,²⁸ and the UV light that larvae are using to detect prey constitutes a tiny percentage of the light in their natural habitat.¹⁵ Thus, a key function of binocularity in this system could be to enhance the detectability of small UV stimuli. The larval brain could implement binocular summation by having inputs from the two eyes interact synergistically to trigger strikes, reducing noise and increasing the system's sensitivity.

Our data suggest that the prey capture strike is best evoked by a bright UV dot

with minimal UV background that is visible to both eyes. Intensity and contrast play a key role in distance estimation for UV-bright prey, but we cannot rule out the involvement of binocular distance cues, particularly in other visual contexts. We propose that zebrafish larvae have evolved visual circuits that are optimally tuned to detect bright UV dots in the binocular visual field and release the strike behavior when these conditions have aligned to maximize the chance of successful prey capture.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2023.06.046>.

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AUTHOR CONTRIBUTIONS

J.L.S., B.K., and Y.L. designed the study, with input from all authors; B.K. and O.J. performed the experiments, data collection, and data pre-processing; B.K. and O.J. did the analysis; B.K. did the visualization; I.P.L. and B.K. designed and built the LightCrafter setup; Y.Z. and V.D.H. helped in experiment preparation and data collection; P.Z. made the arena where experiments was conducted. G.T. provided conceptual input, feedback, and suggestions; J.L.S. wrote the original draft with input from all authors; all authors reviewed and edited the manuscript; J.L.S. acquired the funding.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Invitrogen Ultrapure Low Melting Point Agarose	ThermoFisher Scientific	Cat#16520050
Hepes	Sigma-Aldrich	H4034
Sodium chloride	VMR Chemicals	Cat#27810.262
Potassium chloride	VMR Chemicals	Cat#26764.298
Magnesium sulphate heptahydrate	VMR Chemicals	Cat#25165.260
Calcium nitrate tetrahydrate	Sigma-Aldrich	Cat#237124
Methylene blue	Sigma-Aldrich	M9140
Deposited data		
Data	Mendeley Data	https://doi.org/10.17632/h5xw3cwmyj.1
Experimental models: Organisms/strains		
Zebrafish: Tg(elavl3:H2B- GCaMP6s)	Vladimirov et al. ²⁹	N/A
Software and algorithms		
Custom tracking behavior analysis code	Github	https://github.com/SemmelhackLab/Strike_Code
Python 2.7	Anaconda2	https://www.anaconda.com/
Python 3.8	Anaconda3	https://www.anaconda.com/
ZebraZoom	N/A	https://github.com/oliviermirat/ZebraZoom
DeepLabCut	N/A	https://github.com/DeepLabCut/DeepLabCut
OpenCV	N/A	https://github.com/opencv/opencv
ImageJ	N/A	https://imagej.nih.gov/ij/
Scikit-learn	N/A	https://imbalanced-learn.org/stable/
Seaborn	N/A	https://seaborn.pydata.org/
SciPy	N/A	https://scipy.org/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Julie Semmelhack (jsemmelhack@ust.hk).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Processed data have been deposited at Mendeley Data and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#).
- All original code has been deposited at Github and is publicly available as of the date of the publication. The link is listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECTS DETAILS

Animals

All procedures were approved by the animal ethics committee of HKUST. Zebrafish: Tg(elavl3:H2B- GCaMP6s²⁹) carrying mutations in the *mitfa* allele (nacre) were used for all experiments. Animals were housed under a standard 14:10 light cycle. Larvae were kept in Danieau's solution, and fed with paramecia (*Paramecium multimicronucleatum*, Carolina Biological Supply Company) on days 5 and 7.

6, 7 or 8 dpf larvae were embedded in 2% low melt agarose on Rinzi Plastic Coverslips (Delta Microscopies) one day before experiments, and the eyes and tail were freed from agarose with a razor blade.

METHOD DETAILS

Behavioral assay

Larvae were kept in Danieau's solution, and fed with paramecia (*Paramecium multimicronucleatum*, Carolina Biological Supply Company) on days 5 and 7. 6, 7 or 8 dpf larvae were embedded in 2% low melt agarose on Rinzi Plastic Coverslips (Delta Microscopies) one day before experiments, and the eyes and tail were freed from agarose with a razor blade.

Experiments were carried out in a dark room with a fixed temperature of 22 degrees. The behavioural chamber was a 60x80x30 mm rectangular arena made from transparent acrylic board. The front end of the arena consisted of a 60x30 mm Teflon fabric screen. The arena was illuminated by a 1.25" x 2.00" infrared light (850 nm, MicroBrite High Intensity Spotlight, Advanced Illumination) on the right side, and the side camera was placed on the left side (BFS-U3-04S2M-CS USB 3.1 FLIR Blackfly S, Monochrome Camera) fitted with a right angle zoom lens (VZM 450i Right Angle Zoom Imaging). A second camera (Photon Focus, MV1-D1312E-160-CL-12), fitted with an objective lens (Sigma 105 mm), was placed under the arena to record tail and eye movements. A second identical infrared light illuminated the larva from below. Both the side and bottom camera recorded at a framerate of 200 frames/second.

At the start of the experiment, the coverslip containing the larva was placed into a 3D printed coverslip holder, which was attached to an XYZ stage (Dovetail slide stage, HK Mechanics). It was then lowered into the arena filled with Danieau's solution. The Light Crafter (LCr) visual stimulator was placed in front of the screen at a distance of 13.5 cm.

Spatial Visual Stimulator

We modified a recently published spatial stimulator design⁹ to allow for the use of only one light crafter unit (DLP® Light Crafter™ 4500 modified by EKB Technologies Ltd) instead of two (Figure S1). The light engine consisted of a high powered 385nm "UV" LED (SOLIS-385C Thorlabs) combined with 415nm "Blue" LED (M415L4 Thorlabs), 470nm "Green" LED (M470L4 Thorlabs), 590nm "Red" LED (M590L4 Thorlabs). The bandpass-dichroic mirror for UV (DMLP425R Thorlabs, F39-370 Analysentechnik) was added in the front right before the collimator, combined with the rest of the bandpass-dichroic mirror for other LEDs (F48-559, F48-550, F47-420, F49-480, F39-587 Analysentechnik).

Visual stimuli

Visual stimuli were designed in Python using PsychoPy.³⁰ The prey stimulus, unless otherwise stated, was a 4° diameter UV (385 nm) dot on a black background, sweeping at a speed of 200°/second horizontally from -70° degree to 70°, at an elevation of 27° degrees. The interstimulus interval was 1 minute, with a recording period of 10 seconds. Stimuli were presented in randomized order.

To measure the attenuation of the stimulus in water, we projected a 50° diameter dot on the screen and measured the intensity with a power meter (Thorlabs, PM100USB). We submerged the power meter sensor (Thorlabs, S130VC 200nm -1100 nm) in the behavioural chamber at the location where the fish would normally be. We calculated the intensity of a 4° prey stimulus assuming power per area is constant. To change the stimulus intensity for the intensity compensation and color ratio experiments, the LED power was adjusted by turning the knob on the driver attached to the UV, Blue, Green, or Red LED, and the power meter was used measure the resulting stimulus.

Eye Tracking and hunting initiation

Eye tracking was performed via a custom python code using the OpenCV python library. An appropriate threshold was set to find three large contours within the fish, corresponding to the two eyes and swim bladder. The heading of the fish was defined by a vector starting in the center of the swim bladder and passing through the midpoint between the eye centroids. The angle of each eye was calculated from the image moments of their contours, and the binocular eye angle distribution was fitted to a kernel density estimation using a Gaussian Kernel. The distributions were predominately bimodal, and the threshold for hunting initiation for each fish was defined as one standard deviation below the second peak in the bimodal distribution.¹⁸ In cases where the vergence angle distribution was not bimodal, a fixed threshold of 50° was used.

Strike Classification

To find motion frames within the side camera videos, we used an adaptive threshold approach. We first calculated the greyscale level difference between sequential frames, and if the average greyscale difference between a given frame and the previous one is greater or equal to the mean + 3 x standard deviation of the distribution of the video's average greyscale difference, we considered it a motion frame. For all motion frames, we identified the moving pixels as those with a magnitude value greater than mean + 3 x standard deviation of the distribution of the video's average greyscale difference. We then used the Farneback optical dense flow algorithm¹¹ to calculate the motion vector of each moving pixel between the identified motion frame and the previous frame. The cardinal directions were defined as: left - 0 / 360°, down - 90°, right - 180°, and up - 270°, with a tolerance of ± 30°. If the mode of the moving pixels was upward, then that frame is classified as a strike frame. The algorithm was compared to the strike classifications of two human annotators who both annotated a dataset consisting of 16 trials from 11 larvae.

Tail kinematics Analysis

Tail tracking was done by using the ZebraZoom software.³¹ The ZebraZoom configuration was used to annotate the tail tip and base for each of the videos. The ZebraZoom algorithm finds 10 points between tail tip and tail base and fits it to detect the tail position, shape and tail angle as well as the individual position of each point of the tail in each frame of the video.

Head Elevation Tracking and Calculation

Eye tracking was done on python using open CV. To prepare for the image processing, gaussian blur is first applied. The contour needs to be sorted to identify the eye and the swim bladder as the darkest regions. By sorting the contours in reverse order, we get the eye and swim-bladder as the desired contour. The open cv minimum enclosing circle function is used to find the circle around the eye contour and swim bladder to get the eye coordinates and swim bladder coordinates. Since the eyes and swim bladder can move across different frames of the video, contour identification for eye and swim bladder detection is done for each frame for the algorithm to be dynamic. The head elevation angle is calculated by finding the angle between the three points: midpoint of the swim bladder and midpoint of the eye in the first frame and midpoint of the eye of the corresponding frame.

Jaw Tracking and Jaw Depression Calculation

To identify and quantify jaw depression, deeplabcut³² was used to train a model that can find the mid-point of the eye as well as the ventral surface of the skin below the midpoint of the eye which corresponds to the hyoid bone. The model was trained on a dataset of 40 videos of different lighting conditions. The deeplabcut algorithm identified 19-20 motion frames per video based on the K-means clustering algorithm and these points were manually annotated. The model was trained on 50,000 iterations. The training loss was 0.018. To quantify the jaw depression, we calculated the Euclidean distance between the mid-point of the eye and bottom most part of skin of fish mouth right below the midpoint of the eye for each frame. The derivative of the distance was taken to find the rate of change of jaw movement in each frame. To identify the threshold that can be defined as the “jaw depression”, we took a data set of 42 trials that had putative strikes (head elevations) and 40 without. Then we plotted the maximum derivative value for the these trial and defined a value of 1.83 as the jaw depression threshold (Figure S2A).

Inverse square law

The inverse square law was used to calculate power at different distances by first taking the value at 1 mm to be 35 nW and using the formula¹³:

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

Spectral ratio calculation

Our spectral ratio was calculated based on 4 distances which were 1mm, 5mm, 10mm and 15mm. The absorbance rate was obtained from measurements of river water.³³ The transmission values were found to be 71% for UV, 80% for blue, 90% for green and 98% for red, which is similar to the values we measured in our chamber. Using these values of transmission rate, calculation of the ratios was done by taking the starting value as 1 UV:2 blue at 1mm, based on the natural abundance of UV and blue in water.¹⁵ At 10 mm, based on the transmission rate, the ratio would correspond to 0.8 UV:2 blue. To calculate 5mm and 15mm, the ratio difference from 1mm and 10mm was taken and divided according to the distances, making the ratios 0.9:2 UV: blue at 5mm and 0.7:2 UV: blue at 15mm. The blue intensity was kept constant at all ratios.

Identifying forward swims strike and suction strikes

The forward swims were identified manually based on the known “burst and coast” tail kinematics via the bottom camera, and the side camera video was then used along with the strike detection code to determine whether the forward swim bouts had strikes or not. Similarly, strikes were identified if the strike code detected a strike frame, and no tail movement was detected.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical tests were performed using the SciPy library in Python. All the data was first tested to check if they are from a normal distribution or not using the Kolmogorov Smirnov test. If the p value of this test was significant it was concluded that the data don't follow a normal distribution. Since all our data didn't follow the normal distribution, then for the comparison across different groups the Kruskal-Wallis test (non-parametric Anova test) was conducted. If the pvalue of this test was significant, then we used the Mann-Whitney U (non-parametric test) for doing the pairwise comparison across these groups to find the significantly different pairs.

P value <= 0.0001 was denoted by “*****”

P value <= 0.001 was denoted by “****”

P value <= 0.01 was denoted by “***”

P value <= 0.05 was denoted by “**”

For the linear regression in Figures 5 and 6, we used the linear regression model from statsmodel in Python to find the best fit line.

The different prey coloured stimulus in [Figure 3](#), had 10 trials per wavelength, making a total of 40 trials per fish, with 8 fish in total. All conditions were randomized.

The 1.5mm, 7mm, and 12mm distance experiment in [Figure 4](#), had 10 trials for each distance, adding up to a total of 30 trials per fish, 8 fish in total. All the distances were randomized. The intensity compensation experiment for these distances also had 10 trials per distance, 9 fish in total and all distances were randomized as well.

The inverse square law distance and compensation experiments in [Figure 5](#), had 15 trials for each distance: 1mm, 2mm and 3mm, for a total of 45 trials/fish, 9 fish in total for both experiments. The distances were randomized in both experiments.

The UV prey:background experiment with varying prey intensities in [Figure 6](#), had 7 trials per condition, for a total of 42 trials per fish, 7 fish in total. The UV prey:green background experiment with varying prey intensities, had 7 trials per condition, for a total of 28 trials per fish, 7 fish in total. All the conditions were randomized in both background experiments.

The optimal size stimulus experiment had 7 trials per condition in [Figure S3](#), making a total of 49 trials per fish, 4 fish in total and each trial was randomized with one condition. The optimal speed stimulus had 7 trials per condition, making a total of 56 trials per fish, 4 fish in total and each trial was randomized with one condition.

The constant size control for distance experiment in [Figure S4](#), followed the same procedure as the 1.5mm, 7mm and 12mm distance experiment, also with 8 fish in total.

The UV prey:red and blue background experiment with varying prey intensities in [Figure S6](#), followed the same procedure as the UV prey: green background experiment. The spectral ratio experiment consisted of 8 trials per condition, for a total of 32 trials per fish, and all conditions were randomized, 10 fish in total.

No fish were excluded.